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When Model Complexity Matters in Cross-Subject Naturalistic fMRI Prediction

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Predicting fMRI responses to naturalistic stimuli increasingly relies on pretrained stimulus encoders and complex downstream models. Yet the predictive value of this added complexity remains unclear. We examine the contributions of stimulus representations and downstream nonlinearity, and how their value depends on access to recent measured BOLD history.

00 A shared cross-subject model comparison

Three input settings

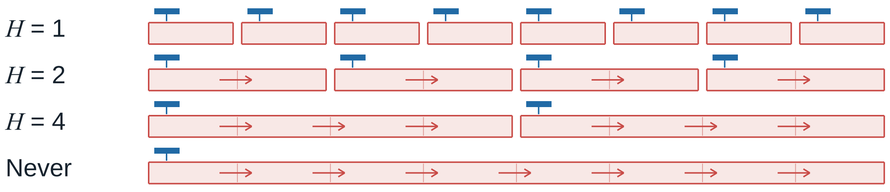
- **Stimulus only**
  Stimulus features; no BOLD history.
- **History only**
  Recent BOLD; no stimulus features.
- **History + stimulus**
  Recent BOLD and stimulus features.

Where is the complexity?

Component	RR-stim	TRIBE v2	MIRAGE	RR-AR	MLP	GRU	BrainVista	NeuroWorld
Stimulus encoder	✓	✓	✓	✓	✓	✓	✓	✓
Mapper from stimulus features to fMRI	L	✓	✓	×	×	×	×	×
Mapper from fMRI history + stimulus features to fMRI	×	×	×	L	✓	✓	✓	✓

✓ = neural network; L = linear; × = unused.

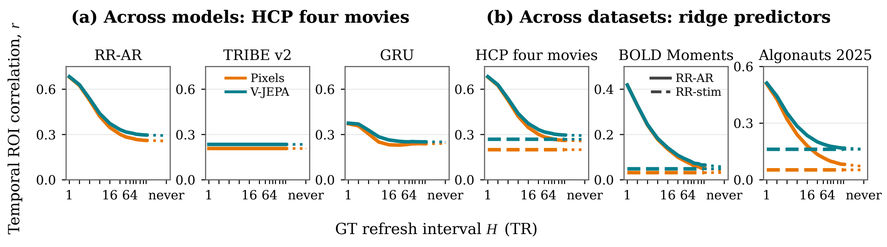
H = steps between measured-history updates. Blue: measured BOLD history. Red: prediction blocks (8 target frames shown).



Evaluation: concatenate predicted blocks and compute temporal Pearson r with measured BOLD. Held-out subject testing: HCP (154 subjects), BOLD Moments (10 subjects), Algonauts (4 subjects).

01 Stronger stimulus representations help

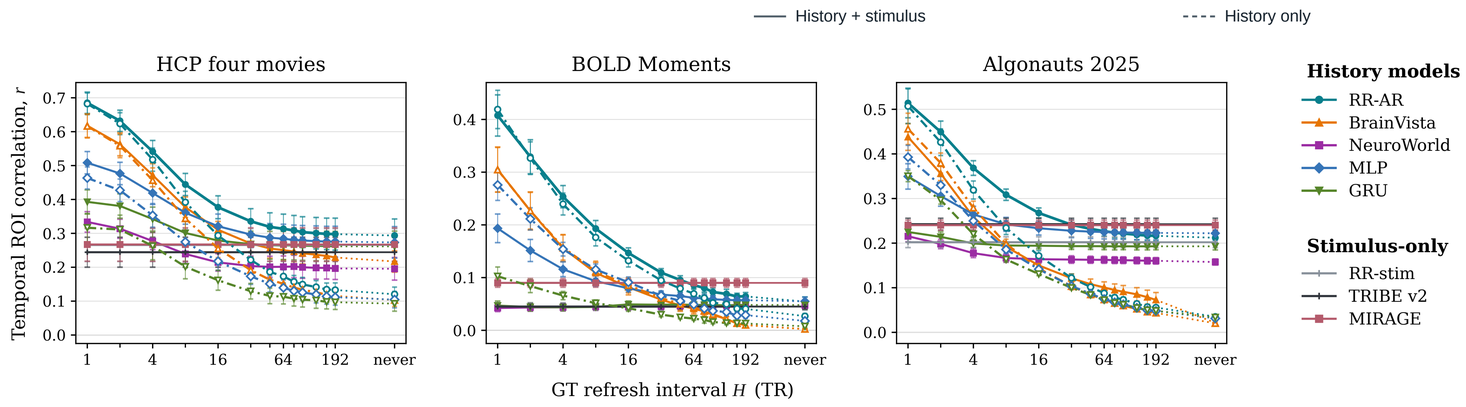
We compare raw pixels with pretrained V-JEPA visual features across predictors and datasets.



V-JEPA generally improves prediction, with the RR-AR advantage becoming clearer as measured-BOLD refreshes become less frequent. Stronger stimulus representations therefore help BOLD prediction.

02 History availability changes the value of nonlinearity

We compare eight predictors and history-only controls across three datasets, varying refresh interval H. Models start with 8 measured frames; "never" means no further refresh. Error bars show SD across held-out subjects.

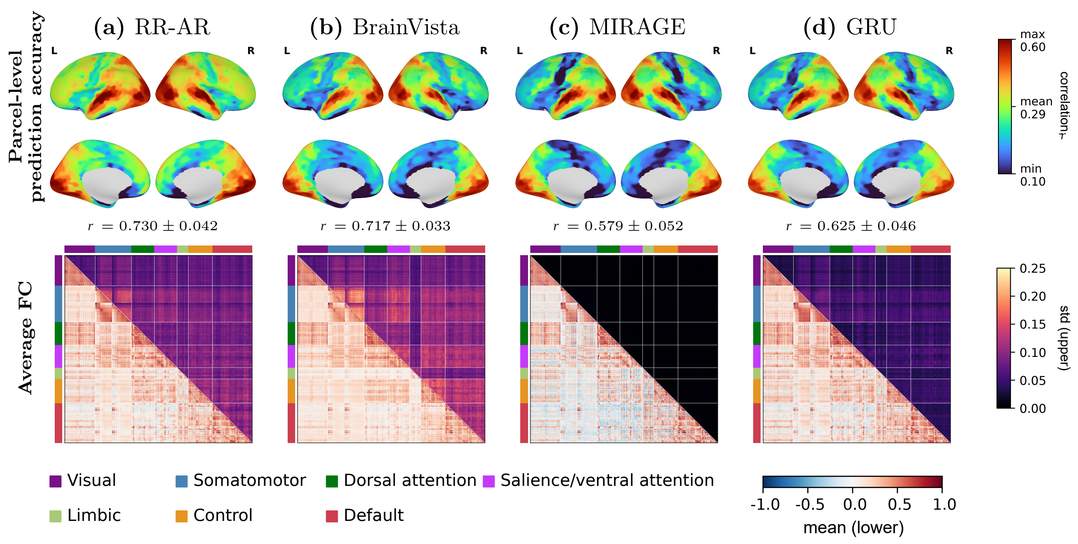


At $H \leq 4$, RR-AR, BrainVista and MLP outperform all stimulus-only references across datasets. History-only RR-AR nearly matches the full model, showing that recent BOLD supplies much of the short-interval predictive signal.

As H increases, history-only RR-AR accuracy declines. Stimulus input adds larger gains on HCP and Algonauts than on BOLD Moments. Measured history and stimulus therefore provide complementary predictive information.

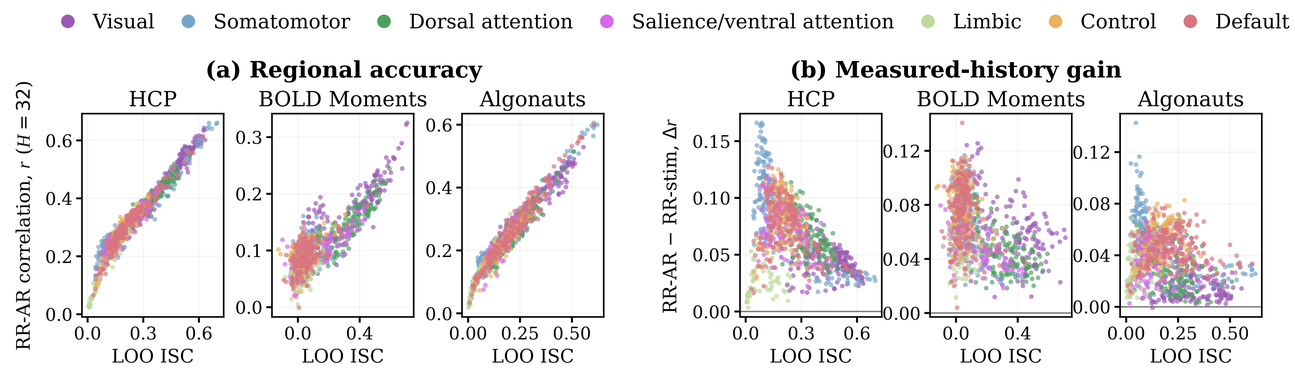
MIRAGE improves on RR-stim on BOLD Moments and Algonauts, while RR-AR generally leads history-conditioned models, including parameter-matched MLP and GRU. The value of nonlinearity depends on history availability.

03 Regional and network-level prediction



On HCP at $H = 32$, maps show temporal prediction correlations; FC matrices show mean FC (lower) and intersubject SD (upper). The models predict visual and auditory cortex comparatively well; RR-AR retains higher correlations in association cortex, where BrainVista also outperforms MIRAGE and GRU. This pattern suggests that recent BOLD may supply accumulated stimulus context and ongoing response information missed by stimulus features alone. History-conditioned models show intersubject FC variability; stimulus-only models show little to none.

04 History helps where responses differ across subjects



Each point is a cortical parcel at $H = 32$. The x-axis shows response consistency: how similarly training subjects respond to the same stimulus. For each parcel, we correlate each subject's standardized response with the mean standardized response of the other training subjects, then aggregate across subjects and matched stimulus groups. The y-axis shows held-out RR-AR prediction correlation (left) or its gain over RR-stim (right). More consistent responses are generally easier to predict. Control and Default networks gain more from history than visual cortex: on HCP, gains are 0.0445 and 0.0417 versus 0.0226, respectively. This suggests that recent BOLD is especially valuable where shared stimulus features provide a less complete account of the response.

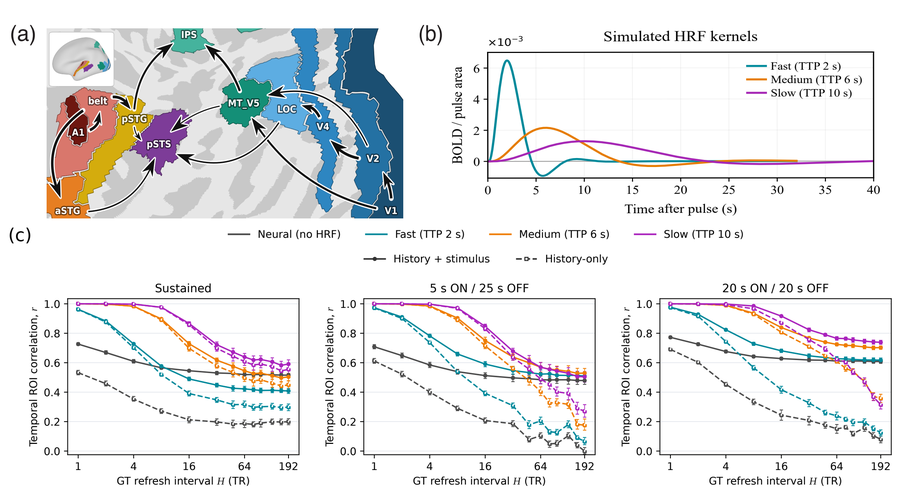
Lab Author Paper

05 Hemodynamic filtering supports linear predictability

Paired simulations vary HRF timing while holding neural trajectories and stimulus inputs fixed. Panels show the circuit (a), HRF kernels (b), and predictions across three stimulus schedules (c). Error bars: 95% bootstrap CIs.

Medium and slow HRFs produce a short-interval plateau, also retained by history-only prediction. Slower HRFs generally extend predictability; stimulus inputs add value at longer intervals.

With neural trajectories held fixed, the simulations identify hemodynamic filtering as a contributor to sustained linear prediction.



Takeaways

- Stronger pretrained stimulus encoders generally improve prediction.
- With history, linear RR-AR is strong in subject-averaged r and association-cortex prediction, and produces intersubject FC variability.
- Adding history yields larger gains in less consistent parcels; Control and Default benefit more than visual cortex.
- HRF filtering helps make linear prediction a strong baseline by sustaining predictability from recent BOLD.

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